

Lecture 5: Evidence worksheet

Name: _____ Group: _____ Assigned case: A / B / C

Opening votes

Editor pitch	Initial verdict and missing evidence
A. Cholesterol is dangerous	
B. Raising HDL protects the heart	
C. Sugar replaces cholesterol as culprit	

Case A: HPS randomized trial

Approximately five years. Outcome: first major vascular event. Each affected participant contributes once to the composite.

Assigned arm	Participants n	With event e
Simvastatin	10,269	2,033
Placebo	10,267	2,585

1. Calculate event risks, ARR (control minus treatment), RR (treatment/control) and RRR. State units. Translate ARR into fewer affected people per 1,000.
2. $SE(ARR) = \sqrt{p_C(1-p_C)/n_C + p_T(1-p_T)/n_T}$. Calculate an approximate 95% interval using $ARR \pm 1.96 SE$.
3. A bar chart starts at 15%. Compare $(p_C - 15)/(p_T - 15)$, using percent units, with p_C/p_T . Specify a repaired encoding and caption.

Calculations and chart specification:

Source: HPS Collaborative Group (2002), Figures 3 and 5. Full links and study scope: [sources.md](#) and [study_dictionary.md](#).

Case B: AIM-HIGH randomized add-on

Both groups receive intensive statin therapy. Mean follow-up: three years. Outcome: first primary composite cardiovascular endpoint.

Assigned add-on	Participants n	With event e
Niacin	1,718	282
Placebo	1,696	274

Niacin-arm median HDL-C: 35 mg/dL at baseline, 42 mg/dL at two years. Reported clinical-outcome HR: 1.02 (95% CI 0.87-1.21).

1. Calculate the change in the HDL median as a percentage. Explain exactly what it measures.

2. Calculate both event risks and the count-based RR.

3. $SE(\log RR) = \sqrt{1/e_T - 1/n_T + 1/e_C - 1/n_C}$. Form $\log RR \pm 1.96 SE$, then exponentiate both limits. Compare with the reported HR interval.

4. Explain why an interval containing 1 does not establish exact equivalence.

5. Sketch separate biomarker/outcome panels or a clinical ratio interval. State axis units, reference, uncertainty and caption.

Source: AIM-HIGH Investigators (2011), original report. Biomarker summaries compare within-arm medians, not paired participant changes. Source link: [sources.md](#).

Case C: NHANES observational cohort

Outcome: CVD mortality. Exposure: modeled usual added-sugar percentage of energy. Q1 midvalue 7.4% is the reference.

Group	Exposure	Demographic HR (95% CI)	Full HR (95% CI)
Q1	7.4%	1 reference	1 reference
Q2	11.4%	1.09 (1.05-1.13)	1.07 (1.02-1.12)
Q3	14.8%	1.23 (1.12-1.34)	1.18 (1.06-1.31)
Q4	18.7%	1.49 (1.24-1.78)	1.38 (1.11-1.70)
Q5	25.2%	2.43 (1.63-3.62)	2.03 (1.26-3.27)

Demographic adjustment: age, sex, race/ethnicity. Full adjustment adds education, smoking, alcohol, physical activity, family CVD history, antihypertensive medication, Healthy Eating Index, BMI, systolic blood pressure, total serum cholesterol and total calories.

1. Q5 begins at 21.3%. Explain why its displayed 25.2% differs from that threshold. State the full-model Q5 comparison and interval.

2. Can HR 2.03 give an absolute mortality probability? Why does model adjustment alone fail to establish causality?

3. Sketch all five estimates at numeric exposure positions. Use a log HR axis, reference 1 and published intervals. Give Q1 a reference marker without an invented CI.

4. Name two omissions in a pitch that shows only Q1/Q5 with no intervals. Explain why this association does not replace the HPS treatment finding.

Source: Yang et al. (2014), Table 2. Both models are adjusted. No participant records or new Cox models are reconstructed. Source link: [sources.md](#).

Workshop and individual exit ticket

Twenty-minute group workshop

5 min calculate and identify omission. 7 min sketch a repaired figure. 5 min write an evidence brief. 3 min respond to a reviewer.

Figure specification: question, quantity, encoding, axes/reference, uncertainty and caption.

Three-sentence evidence brief

1. Finding:

2. Numerical support:

3. Study scope:

Reviewer challenge and response:

Individual exit ticket

1. Translate HPS absolute benefit into affected people per 1,000, including the period.

2. Explain why improved HDL alone does not establish protection.

3. Explain why the sugar association does not acquit LDL.

4. Give the null reference for a difference and a ratio.
